

Examiner
only

(b) (i) The efficiency of energy transfer between one trophic level and the next is **10%**. Calculate the percentage of the energy entering trophic level **1** that reaches trophic levels **3, 4** and **5**. **Write your answers in the pyramid in Image 3.1.** [2]
Space for working.

(ii) Suggest why the food chain represented by the pyramid of biomass in **Image 3.1** could not sustain a trophic level **6**. [1]

.....

.....

(iii) State **one** way in which energy is lost from a food chain. [1]

.....

(c) Underline one correct statement about pyramids of biomass from the following list. [1]

As you go up most pyramids of biomass from one trophic level to the next the:

- size of the organisms decreases and their numbers increase.
- size of the organisms decreases and their numbers decrease.
- size of the organisms increases and their numbers increase.
- size of the organisms increases and their numbers decrease.

3430UA01
11

8



Examiner
only

5. Describe the pathway taken by the blood in a double circulatory system. Start your account with the return of blood along the pulmonary vein from the lungs to the left atrium of the heart. Your account must include details of how backflow of blood is prevented. [6 QER]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

6

